Supporting Information

Metabolism of a Representative Oxygenated Polycyclic Aromatic Hydrocarbon (PAH) Phenanthrene-9,10-quinone in Human Hepatoma (HepG2) Cells

Meng Huang, [†] Li Zhang, [†] Clementina Mesaros, [‡] Suhong Zhang, [‡] Michael A. Blaha, [†] Ian A. Blair, [†,‡] and Trevor M. Penning [*,†,‡]

[†] Center of Excellence in Environmental Toxicology and [‡] Center for Cancer Pharmacology, Department of Pharmacology, Perelman School of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania 19104-6160, United States

* To whom correspondence should be addressed: Department of Pharmacology, Perelman School of Medicine, University of Pennsylvania, 1315 Biomedical Research Building II/III, 421 Curie Boulevard, Philadelphia, Pennsylvania 19104-6160. Telephone: (215) 898-9445. Fax: (215) 573-0200. Email: penning@upenn.edu

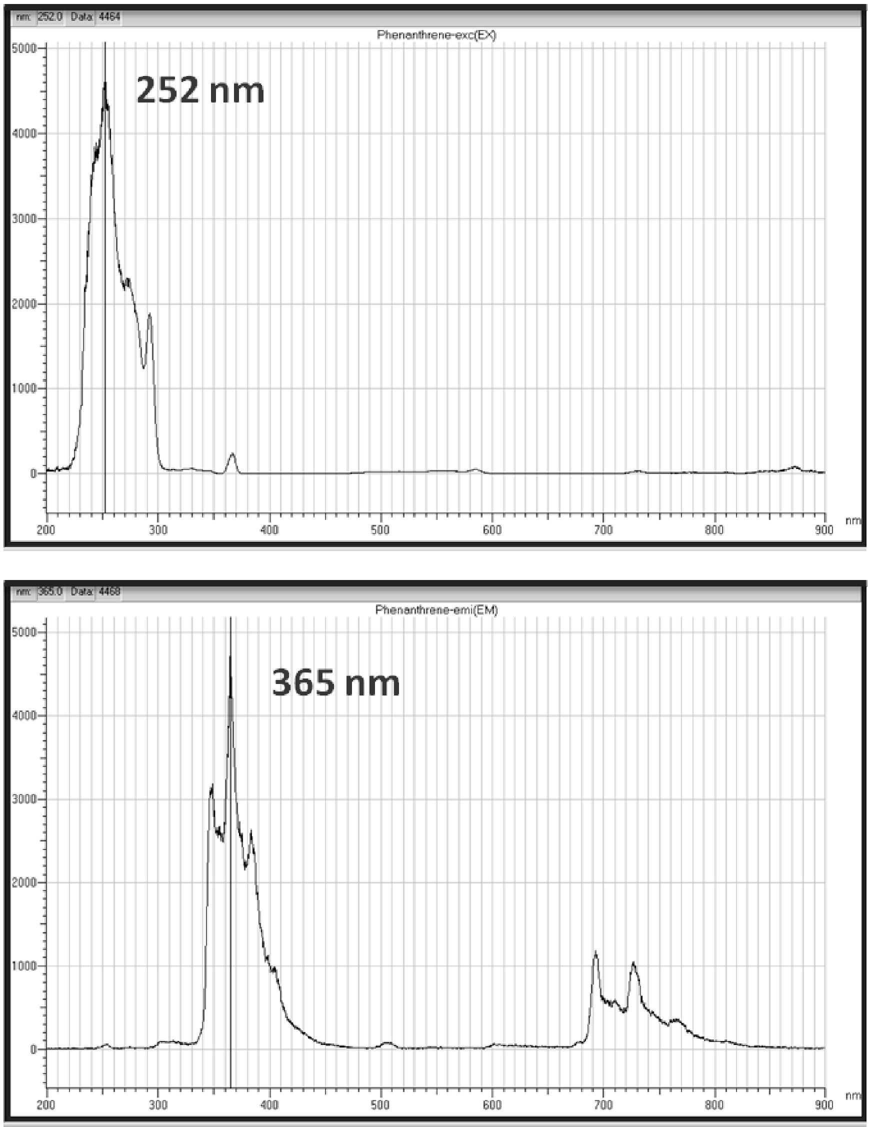

Figure S1. Excitation wavelength and emission wavelength spectra of phenanthrene standard in ethanol (1 μg/mL).

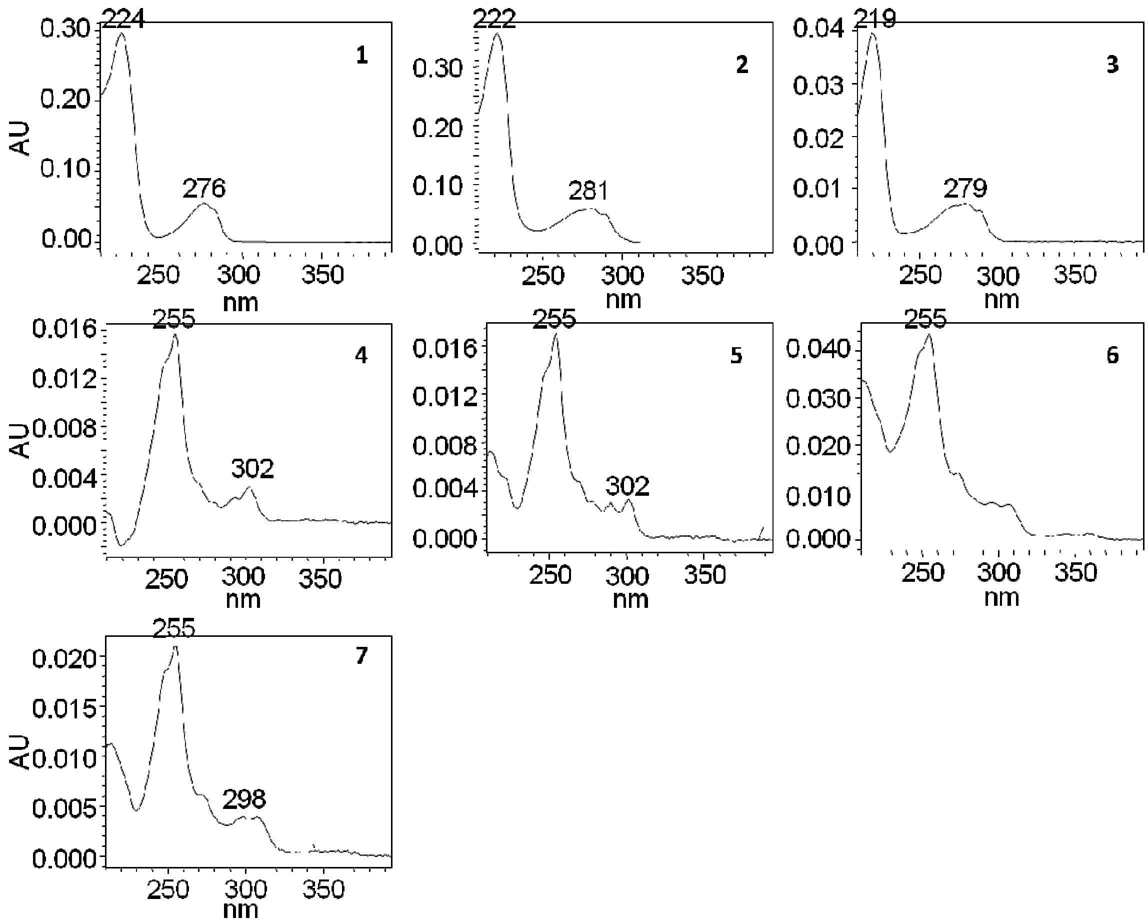

Figure S2. UV spectra of metabolite 1-7 from organic phase of media in HepG2 cells following treatment with 1 μM phenanthrene-9,10-quinone for 24 h.

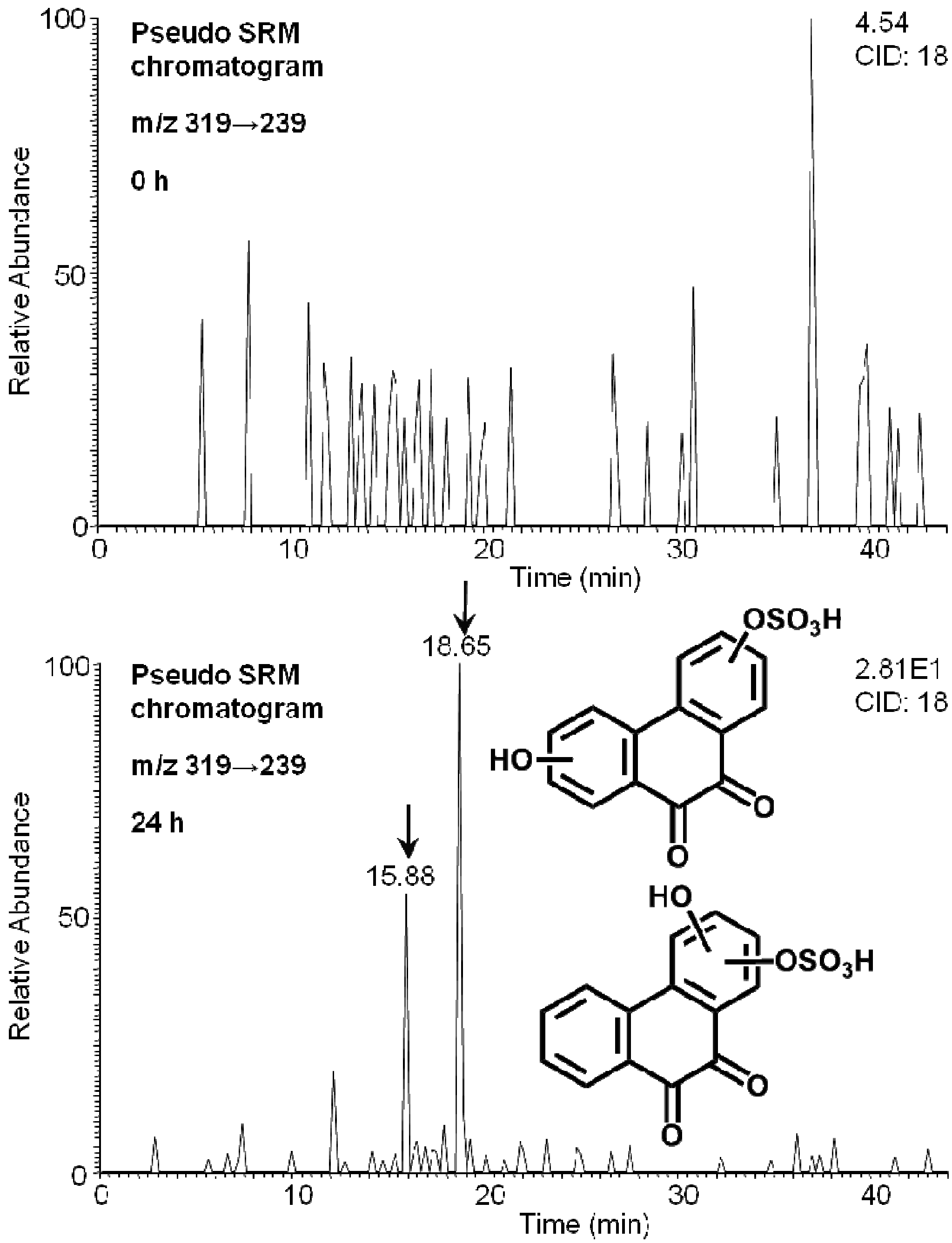

Figure S3. Extracted ion chromatograms of pseudo SRM transitions for *O-mono*-sulfonated *bis*-phenols of phenanthrene-9,10-quinone detected in HepG2 cells at 0 h and 24 h in the negative ion mode.

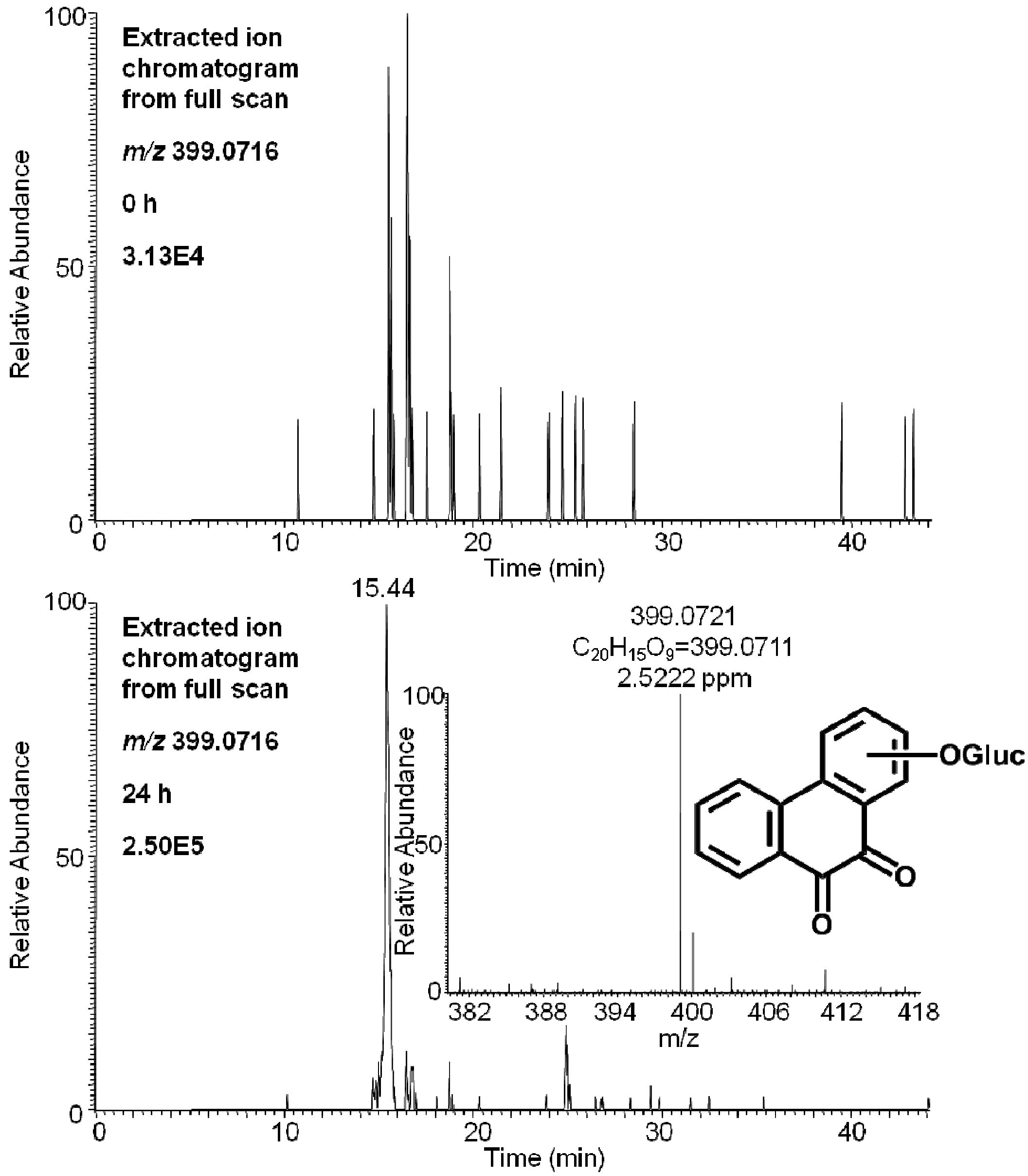

Figure S4. Extracted ion chromatograms of Orbitrap full scan for *mono*-hydroxylated-*O*-*mono*-glucuronosyl-phenanthrene-9,10-quinone detected in HepG2 cells at 0 h and 24 h in the negative ion mode.

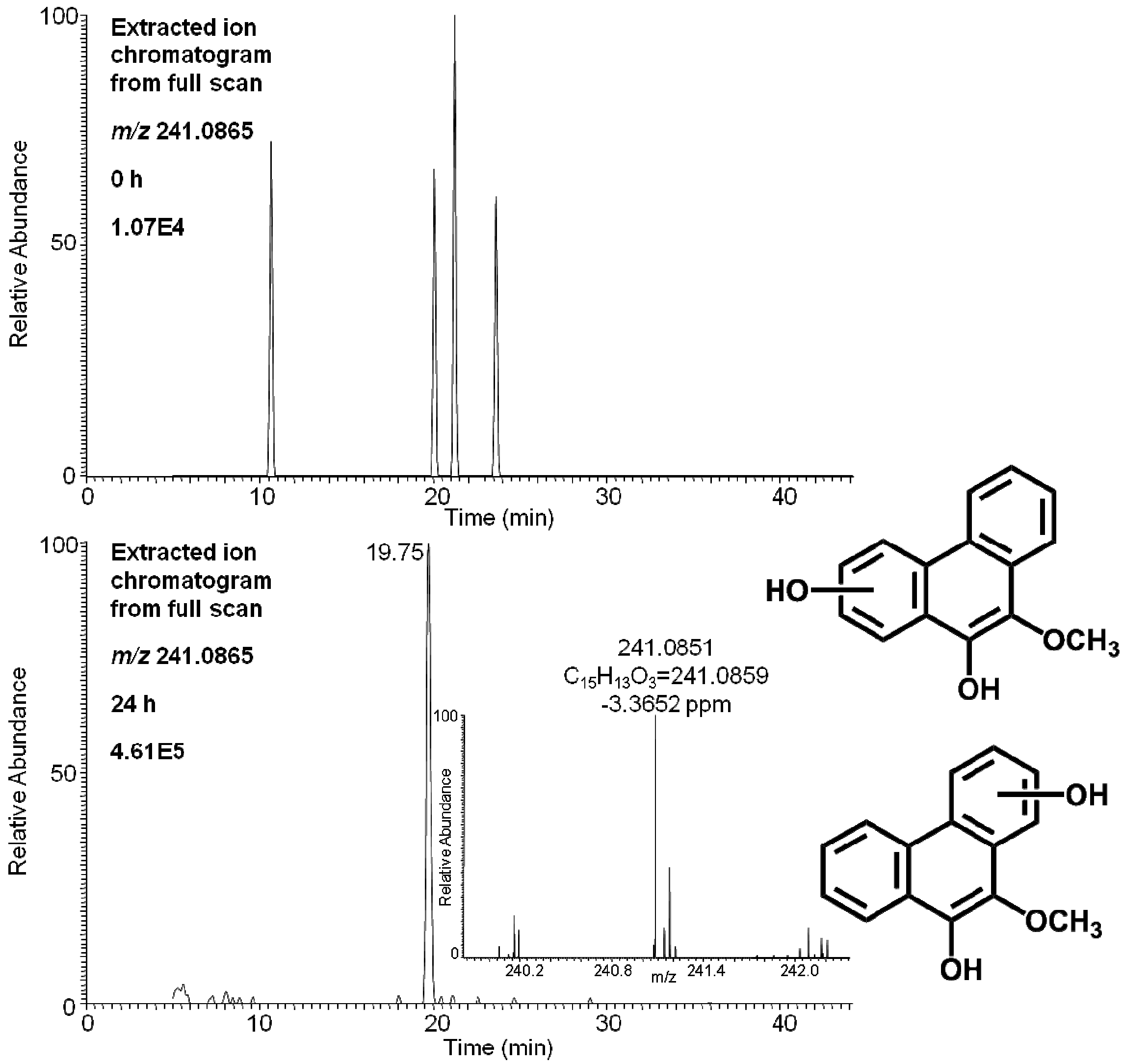

Figure S5. Extracted ion chromatograms of Orbitrap full scan for *mono*-hydroxylated-*O*-*mono*-methyl-phenanthrene-9,10-catechol detected in HepG2 cells at 0 h and 24 h in the positive ion mode.

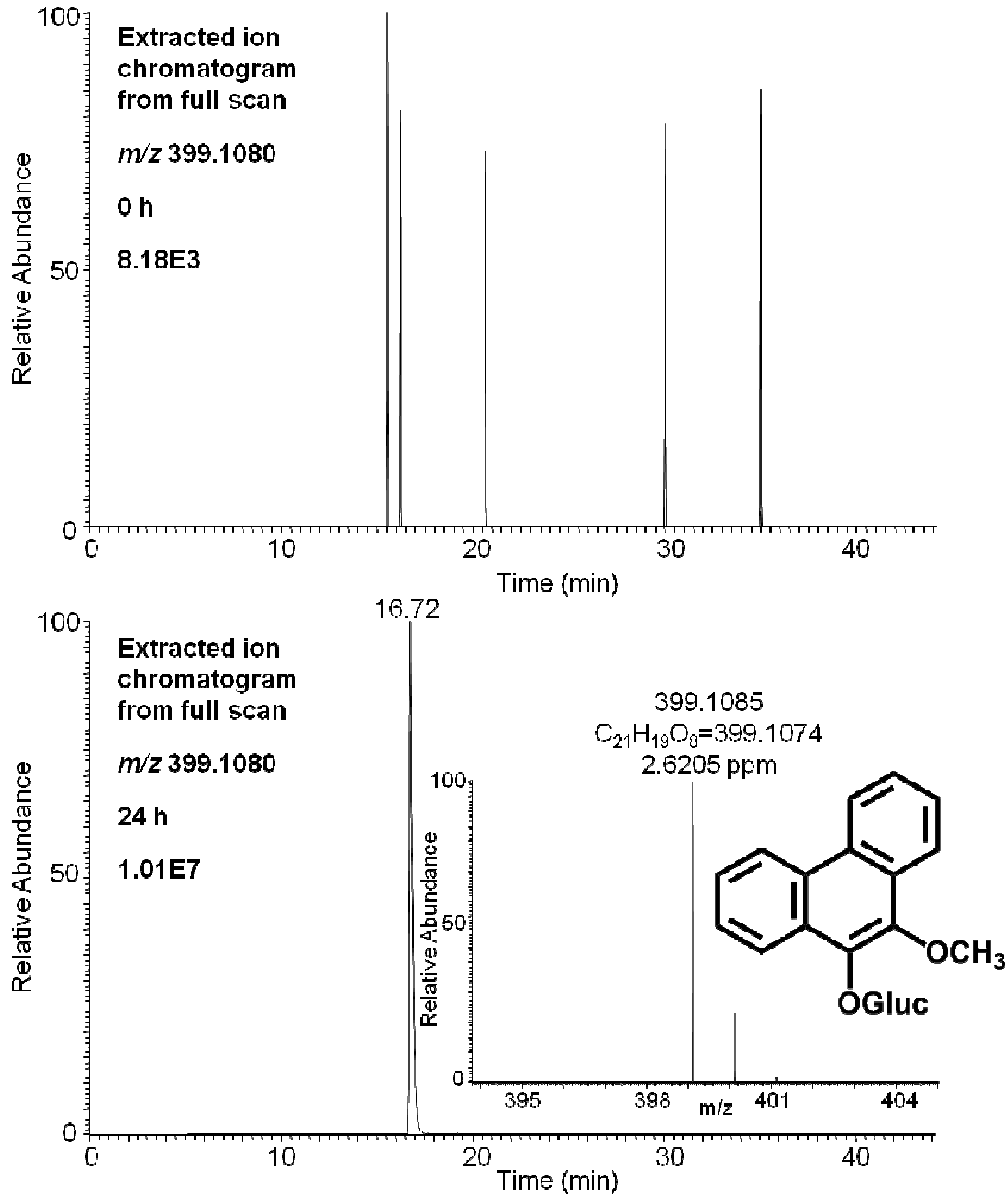

Figure S6. Extracted ion chromatograms of Orbitrap full scan for *O-mono*-methyl-*O-mono*-glucuronosyl-phenanthrene-9,10-catechol detected in HepG2 cells at 0 h and 24 h in the negative ion mode.

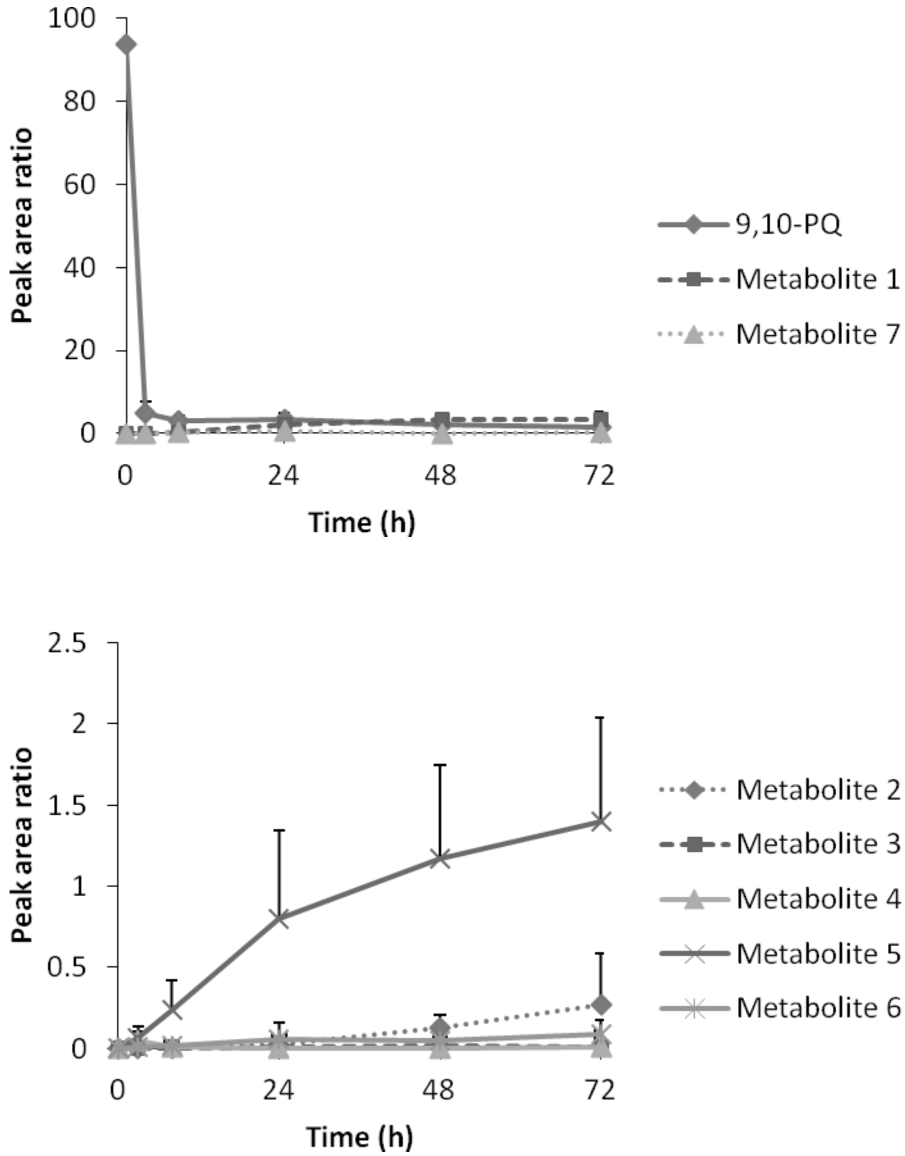

Figure S7. Time course of phenanthrene-9,10-quinone (9,10-PQ) consumption and its metabolite formation in HepG2 cells at 0 h, 3h, 8 h, 24 h, 48 h and 72 h. The quantitation of phenanthrene-9,10-quinone, metabolite 1 and 7 was based on the UV peak area ratio of the analyte and phenanthrene (internal standard). The quantitation of metabolite 2-6 was based on the fluorescence peak area ratio of the analyte and phenanthrene (internal standard).